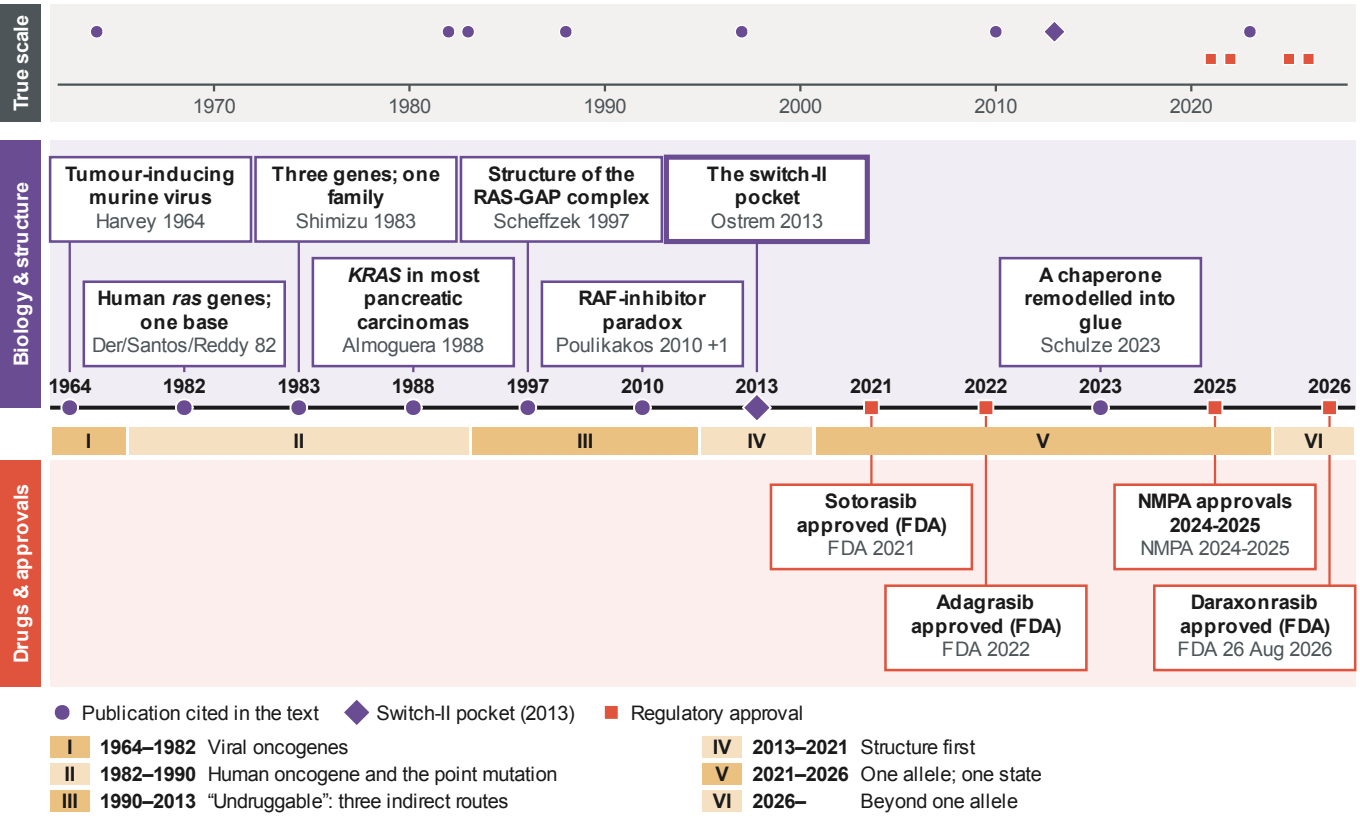
True scale
1970
1980
1990
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2010
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Biology & structure
Tumour-inducing murine virus
Harvey 1964
Three genes; one family
Shimizu 1983
Structure of the RAS-GAP complex
Scheffzek 1997
The switch-II pocket
Ostrem 2013
Human ras genes; one base
Der/Santos/Reddy 82
KRAS in most pancreatic carcinomas
Almoguera 1988
RAF-inhibitor paradox
Poulikakos 2010 +1
A chaperone remodelled into glue
Schulze 2023
1964
1982
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1997
2010
2013
2021
2022
2023
2025
2026
I
II
III
IV
V
VI
Drugs & approvals
Sotorasib approved (FDA)
FDA 2021
Adagrasib approved (FDA)
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NMPA approvals 2024-2025
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Switch-II pocket (2013)
Regulatory approval
I 1964–1982 Viral oncogenes
II 1982–1990 Human oncogene and the point mutation
III 1990–2013 "Undruggable": three indirect routes
IV 2013–2021 Structure first
V 2021–2026 One allele; one state
VI 2026– Beyond one allele